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Maximal Weakly--Intuitionistic Logics

Abstract. This article introduces the three-valued *weakly-intuitionistic* logic I^1as a counterpart of the *paraconsistent* calculus P^1 studied in [11]. I^1 is shown to be complete with respect to certainthree-valued matrices. We also show that in the sense that any proper extension of I^1 collapses to classical logic.

The second part shows that I^1 is algebraizable in the sense of Block and Pigozzi (cf. [2]) in a way very similar to the algebraization of P^1 given in [8].

In the last part of the paper we suggest the definition of certain hierarchies of finite-valued propositional paraconsistent and weakly-intuitionistic calculi, and comment on their intrinsic interest.

1. Motivations

The three-valued paraconsistent system P^1 was introduced in [11] with the purpose of presenting the simplest possible paraconsistent logic. This system is the simplest paraconsistent logic in the sense that it is a proper subsystem of the classical propositional calculus (PC), and adding to the axioms of P^1 any classical tautology (which is not a P^1-theorem) one obtains PC.This maximality property of P^1 makes it the closest possible to PC (of course, we do not claim that P^1 is the only system having such property).

From the semantic point of view, the simplicity of P^1 is characterized by the following property: if some P^1 -theory $T$is inconsistent, that is, if $T \models \alpha$ and $T \models \neg\alpha$ then α must be an atomic formula; in other words, inconsistencies occur in P^1 theories only at the atomic level.

The main idea behind the formulation of P^1 is to add a new truth-value T^* to the classical boolean values $\{T, F\}$ in such a way as to introduce a *perturbation* in classical trueness: the truth-values T andT^* differ with respect to negation only, and coincide otherwise, i.e., $T^*$and $\neg T^*$ are both true. Intuitively, thus, the paraconsistent calculus P^1 can be seen as *truth-balanced* in the sense that it contains more "true" than "false" logic values.

It seems quite natural, on the other hand, to add a truth-value F^* to the boolean values (instead of T^*) with dual meaning, that is,to introduce a perturbation in classical falsity in such a way that F^* and $\neg F^*$ are both false; in this form we obtain the basic formulation for a *falsity-balanced*

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calculus I^1 (also a proper subsystem ofPC) whose properties are investigated in this work. We define a calculus (whose language contains $\vee$) to be *weakly-intuitionistic* if the formula $\alpha \vee \neg\alpha$ is not valid (i.e. the law of excluded middle cannot be proved); in this sense, I^1 is a weakly-intuitionistic logic, as shown in Section 5.

The main characteristics of I^1 are that the law of excluded middle is false for atomic formulas only, and that I^1 is maximal with respect to PC.Defining a calculus (whose language contains $\wedge$) to be *weakly-paraconsistent* if the formula $\alpha \wedge \neg\alpha$ is not valid (i.e. if the law of non-contradiction cannot be proved) we can see weakly-paraconsistent and weakly-intuitionistic logics as dual. In our case, the duality between I^1 and P^1 is even stricter, due to the maximality of both systems with respect to PC and to the fact that the (respective) laws of excluded middle and non–contradiction are invalid at atomic level only. A broader study of the duality between paraconsistent and intuitionistic systems will be presented by E. H. Alves and the first author in a forthcoming paper. A system similar to I^1 has also been studied from a different perspective in [3] and [7].

We shall give a *constructive* proof of completeness for the system (although systems of this sort can be easily handled through the general method of finitary trees, cf. [4]).This constructive proof simplifies the work on the algebraic aspects of I^1 studied in the second part.

2. Basic definitions and results on I^1

The syntax of I^1 is composed of three classes of symbols:

1. Propositional variables $p_1, p_2, \ldots, p_n, \ldots$;
2. Logic connectives $\rightarrow$ for implication and $\neg$ for negation;
3. Auxiliary symbols (,).

The notions of well-formed formulas, atomic formulas, schemes of formulas, etc, are the usual ones (that is, the same as in classical logic). Greek letters are used as metavariables over formulas.

If ϕ, ψ, θ are formulas, then the following schemes are axioms:

Axiom 1 $\phi \rightarrow (\psi \rightarrow \phi)$

Axiom 2 $(\phi \rightarrow (\psi \rightarrow \theta)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \theta))$

Axiom 3 $(\neg\neg\phi \rightarrow \neg\psi) \rightarrow ((\neg\neg\phi \rightarrow \psi) \rightarrow \neg\phi)$

Axiom 4 $\neg\neg(\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \psi)$

The only rule of inference of I^1 is *Modus Ponens*:

$$\frac{\phi, \phi \to \psi}{\psi}$$

where ϕ, ψ are any formulas.

The notions of proof, theorem and consequence of a set of formulas Γ are the same as in the classical case (or that of P^1 in [11]) and are denoted, in this case, by $\vdash_{I^1} \alpha$ and $\Gamma \vdash_{I^1} \alpha$. It is clear, from the axioms, that I^1 is a subsystem of PC. These four axioms are easily seen to be independent, as can be checked through appropriate (three-valued) matrices which satisfy all of them with one exception.

We prove in the following some elementary properties about I^1 which are necessary for our proof of the Completeness Theorem; the rule of Modus Ponens will be used several times without explicit mention:

PROPOSITION 1. (Deduction Theorem) *For every set of I^1-formulas $\Gamma \cup \{\phi, \psi\}$, the following holds:*

$$\Gamma, \phi \vdash_{I^1} \psi \textit{ if and only if } \Gamma \vdash_{I^1} \phi \to \psi.$$

PROPOSITION 2. *For every I^1-formulas ψ, θ and ϕ, the following holds*:

(i) $\vdash_{I^1} \psi \to \psi$ (Reflexivity).
(ii) $\phi \to \psi, \psi \to \theta \vdash_{I^1} \phi \to \theta$ (Transitivity).
(iii) $(\phi \to \psi) \to \theta \vdash_{I^1} \psi \to \theta$ (Simplification)
(iv) $\phi \to (\psi \to \theta) \vdash_{I^1} \psi \to (\phi \to \theta)$ (Permutation)

The proof of Propositions 1 and 2 is a direct consequence of Axioms 1 and 2.

Though negations can be arbitrarily expanded (as shown in Proposition 5 below) the law of reduction of negations (Proposition 3 below) holds for non-atomic formulas only; similarly, the Law of non-contradiction (Proposition 4) is valid for negated formulas only. Proposition 6 shows that if arbitrary reduction of negations is allowed we obtain the classical forms of the law of non-contradiction and contraposition.

PROPOSITION 3. *If ψ is a non-atomic formula then*

$$\vdash_{I^1} \neg\neg\psi \to \psi.$$

PROOF. If ψ is non-atomic then either ψ is $\phi \to \theta$ or ψ is $\neg\phi$.

In the first case, Axiom 4 obtains $\vdash_{I^1} \neg\neg(\phi \to \theta) \to (\phi \to \theta)$, that is, $\vdash_{I^1} \neg\neg\psi \to \psi$.

In the second case, the proof runs as follows:

1. $\neg\neg\neg\phi$ (hypothesis)
2. $\neg\neg\neg\phi \to (\neg\neg\phi \to \neg\neg\neg\phi)$ (Axiom 1)
3. $\neg\neg\phi \to \neg\neg\neg\phi$ (from (1) and (2))
4. $\neg\neg\phi \to \neg\neg\phi$ (from Proposition 2.(i))
5. $(\neg\neg\phi \to \neg\neg\neg\phi) \to ((\neg\neg\phi \to \neg\neg\phi) \to \neg\phi)$ (Axiom 3)
6. $\neg\phi$ (from (3),(4), (5))
7. $\vdash_{I^1} \neg\neg(\neg\phi) \to (\neg\phi)$ by the Deduction Theorem, i.e., $\vdash_{I^1} \neg\neg\psi \to \psi$. ■

PROPOSITION 4. (Law of Non-Contradiction, negative form)

$$\vdash_{I^1} (\neg\phi \to \neg\psi) \to ((\neg\phi \to \psi) \to \neg\neg\phi).$$

PROOF.

1. $\neg\phi \to \neg\psi$ (hypothesis)
2. $\neg\phi \to \psi$ (hypothesis)
3. $(\neg\neg\neg\phi \to \neg\psi) \to ((\neg\neg\neg\phi \to \psi) \to \neg\neg\phi)$ (Axiom 3)
4. $\neg\neg\neg\phi \to \neg\phi$ (Proposition 3)
5. $\neg\neg\neg\phi \to \neg\psi$ (from (1) and (4), using Proposition 2(ii))
6. $\neg\neg\neg\phi \to \psi$ (from (2) and (4), using Proposition 2(ii))
7. $\neg\neg\phi$ (from (3),(5) and (6)

and the result follows by the Deduction Theorem. ■

PROPOSITION 5. (Law of expanding negations)

$$\vdash_{I^1} \phi \to \neg\neg\phi.$$

PROOF.

1. $(\neg\phi \to \phi) \to \neg\neg\phi$ (from appropriate forms of Proposition 2.(i) and Proposition 4, Modus Ponens)

2. $\phi \to (\neg\phi \to \phi)$ (Axiom 1)

3. $\vdash_{I^1} \phi \to \neg\neg\phi$ (from (1) and (2) above, and Proposition 2.(ii)). ■

By a *weak law* of I^1 we mean a deduction of a classical propositional law in I^1 from hypothesis of the form $\neg\neg\phi \to \phi$ (for some wff ϕ). The classical laws which hold in I^1 are said to be *strong laws* of I^1.

PROPOSITION 6. *The following weak and strong laws hold in* I^1:

(i) $\neg\neg\phi \to \phi \vdash_{I^1} (\neg\phi \to \neg\psi) \to ((\neg\phi \to \psi) \to \phi)$

(ii) $\neg\neg\phi \to \phi \vdash_{I^1} (\neg\phi \to \neg\psi) \to (\psi \to \phi)$

(iii) $\neg\neg\phi \to \phi \vdash_{I^1} (\phi \to \psi) \to (\neg\psi \to \neg\phi)$

(iv) $\neg\neg\phi \to \phi \vdash_{I^1} (\phi \to (\neg\psi \to \neg(\phi \to \psi)))$

(v) $\neg\neg\phi \to \phi \vdash_{I^1} \neg\psi \to (\psi \to \phi)$

(vi) $\neg\neg\phi \to \phi, \neg\phi \to \psi, \phi \to \psi, \vdash_{I^1} \neg\neg\psi$

(vii) $(((\phi \to \phi) \to \phi) \to \theta) \to (\phi \to \theta)$

(viii) $(\neg((\phi \to \phi) \to \phi) \to ((\phi \to \phi) \to \phi)) \to \phi$

(ix) $\neg\neg\phi \to \phi \vdash_{I^1} \neg\phi \to (\phi \to \psi)$

PROOF. (i) Direct application of Proposition 2.(ii) and Proposition 4.
(ii) Let Σ be the set of formulas $\{\neg\neg\phi \to \phi, \neg\phi \to \neg\psi, \psi\}$. Then:

1. $\Sigma \vdash_{I^1} \neg\phi \to \neg\psi$ (as this formula belongs to Σ)

2. $\Sigma \vdash_{I^1} \neg\phi \to \psi$ (using Axiom 1, since ψ belongs to Σ)

3. $\Sigma \vdash_{I^1} \phi$ by item (i) above, and the result follows by the Deduction Theorem.

(iii)

1. $\neg\neg\phi \to \psi$ (hypothesis)

2. $\phi \to \psi$ (hypothesis)

3. $\neg\psi$ (hypothesis)

4. $\neg\neg\phi \to \psi$ (from (1) and (2), Proposition 2.(ii))

5. $\neg\psi \to (\neg\neg\phi \to \neg\psi)$ (Axiom 1)

6. $\neg\neg\phi \to \neg\psi$ (from (3) and (5), Modus Ponens)

7. $(\neg\neg\phi \to \neg\psi) \to ((\neg\neg\phi \to \psi) \to \neg\phi)$ (Axiom 3)

8. $\neg\phi$ (from (4), (6), (7), Modus Ponens)

9. $\neg\phi \to (\neg\psi \to \neg\phi)$ (Axiom 1)

10. $\neg\psi \to \neg\phi$ (from (8), (9), Modus Ponens) and the result follows from hypothesis (1) and (2) and the Deduction Theorem.

(iv) Let $\Sigma = \{\neg\neg\phi \to \phi, \phi, \neg\psi\}$

1. $\Sigma, \neg\neg(\phi \to \psi) \vdash_{I1} \psi$(since $\phi \in \Sigma$ and $\neg\neg(\phi \to \psi) \to (\phi \to \psi)$ holds by Proposition 3)

2. $\Sigma \vdash_{I1} \neg\neg(\phi \to \psi) \to \psi)$ (from (1) and the Deduction Theorem)

3. $\Sigma \vdash_{I1} \neg\neg(\phi \to \psi) \to \neg\psi)$ (since $\neg\psi \in \Sigma$, using Axiom 1)

4. $\Sigma \vdash_{I1} \neg(\phi \to \psi)$(by (2), (3) and Axiom 3).

(v)

1. $\neg\neg\phi \to \phi$ (hypothesis)

2. $\neg\psi$ (hypothesis)

3. ψ (hypothesis)

4. $\neg\phi \to \neg\psi$ (from (2) and Axiom 1)

5. $\neg\phi \to \psi$ (from (3) and Axiom 1)

6. ϕ (from Part (i) above, (1), (4) and (5))

(vi)

1. $\neg\neg\phi \to \phi$ (hypothesis)

2. $\neg\phi \to \psi$ (hypothesis)

3. $\phi \to \psi$ (hypothesis)

4. $\neg\psi \to \neg\phi$ (1,3 and (iii))

5. $\psi \to \neg\neg\psi$ (Proposition 5)

6. $\neg\phi \to \neg\neg\psi$ (2, d Proposition 2.(ii))

7. $\neg\psi \to \phi$ (1,6 and (iii))

8. $\neg\neg\psi \to \phi$ (4,7 and (Proposition 4)

(vii) By the Deduction Theorem, it is sufficient to prove:

$$(\phi \to \phi) \to \phi) \to \theta, \phi \vdash_{\mathrm{I}^1} \theta$$

1. ϕ (hypothesis)

2. $((\phi \to \phi) \to \phi) \to \theta$ (hypothesis)

3. $(\phi \to \phi) \to \phi$ (from (1), Axiom1)

4. θ (from (2) and (3), Modus Ponens)

(viii) Let α be the formula $(\phi \to \phi) \to \phi$; it suffices to show that $\neg\alpha \to \alpha \vdash_{\mathrm{I}^1} \phi$

1. By 6.(vi), holds: $\neg\neg\alpha \to \alpha, \neg\alpha \to \alpha, \alpha \to \alpha \vdash_{\mathrm{I}^1} \neg\neg\alpha$

2. $\neg\neg\alpha \to \alpha$, (Proposition 3, since α is non atomic)

3. $\alpha \to \alpha$ (Proposition 2(i))

4. $\neg\alpha \to \alpha \vdash_{\mathrm{I}^1} \alpha$(from (1), (2) and Proposition 2.(ii))

5. $\phi \to \phi$ (Proposition 2.(i))

6. $\neg\alpha \to \alpha \vdash_{\mathrm{I}^1} \phi$ (from (4), (5) and the definition of α)

(ix)

1. $\phi \to (\neg(\phi \to \psi) \to \phi)$ (Axiom 1)

2. $\neg\neg(\phi \to \psi) \to (\phi \to \psi)$ (Proposition 3)

3. $\phi \vdash_{\mathrm{I}^1} \neg\phi \to \neg\neg(\phi \to \psi)$ (by 6.(v) and (2) above)

4. $\phi \vdash_{I^1} \neg\phi \to (\phi \to \psi)$ (by (3) and (2), transitivity)
5. $\neg\phi \to (\neg(\neg\phi \to \psi) \to \neg\phi)$ (Axiom 1)
6. $\neg\neg(\neg\phi \to \psi) \to (\neg\phi \to \psi)$ (Proposition 3)
7. $\neg\phi \vdash_1 \neg\neg\phi \to \neg\neg(\neg\phi \to \psi)$ (by 6.(v) and (6) above)
8. $\neg\phi \vdash_{I^1} \neg\neg\phi \to (\neg\phi \to \psi)$ (by (7) and (6), transitivity)
9. $\phi \to \neg\neg\phi$ (Proposition 5)
10. $\neg\phi \vdash_{I^1} \phi \to (\neg\phi \to \psi)$ (by (8) and (9), transitivity)
11. $\neg\phi \vdash_{I^1} \neg\phi \to (\phi \to \psi)$ (by (10) and Proposition 2.(iv))
12. $\neg\neg\phi \to \phi \vdash_{I^1} \neg\neg(\neg\phi \to (\phi \to \psi))$(from (4) and (11),through 6 .(vi))
13. $\neg\neg(\neg\phi \to (\phi \to \psi)) \to (\neg\phi \to (\phi \to \psi))$ (Proposition 3)
14. $\neg\neg\phi \to \phi \vdash_{I^1} \neg\phi \to (\phi \to \psi)$ (from (12) and (13),trasnsitivity) ■

The last part of the following result shows that I^1 is not a paraconsistent logic, since we can prove the strong form of the Law of Non-Contradiction; the first items are necessary for proving completeness:

PROPOSITION 7.

(i) (Peirce's Law) $((\phi \to \theta) \to \phi) \to \phi$

(ii) $\neg(\neg\neg\theta \to \theta) \to \neg\neg\theta$

(iii) $\neg(\neg\neg\phi \to \phi) \vdash_{I^1} \neg\phi \to \theta$

(iv) (Full Law of Non-Contradiction) $\vdash_{I^1} \neg\phi \to (\phi \to \theta)$.

PROOF. (i) Let α be the formula $(\phi \to \phi) \to \phi$;

1. $(\phi \to \theta) \to \phi$ (hypothesis)
2. $(\alpha \to \theta) \to (\phi \to \theta)$(from 6.(vii))
3. $(\alpha \to \theta) \to \phi$ (by (1) and (2), transitivity)
4. $\phi \to \alpha$ (Axiom (1))
5. $(\alpha \to \theta) \to \alpha$ (by (1) and (4), transitivity)

6. $\neg\neg\alpha \to \alpha$ (Proposition 3)
7. $\neg\alpha \to (\alpha \to \theta)$ (from (6) and Proposition 6.(ix))
8. $\neg\alpha \to \alpha$ (by (5) and (7),transitivity)
9. $(\neg\alpha \to \alpha) \to \phi$ (from 6.(viii))
10. ϕ (from (8) and (9)).

(ii)

1. $\neg(\neg\neg\theta \to \theta)$ (hypothesis)
2. $\neg\neg\neg\neg\theta \to \neg\neg\theta$ (Proposition 3)
3. $\neg(\neg\neg\theta \to \theta) \to ((\neg\neg\theta \to \theta) \to \neg\neg\theta)$ (from (2) and Proposition 6.(v))
4. $(\neg\neg\theta \to \theta) \to \neg\neg\theta$ (from (1) and (3))
5. $((\neg\neg\theta \to \theta) \to \neg\neg\theta) \to \neg\neg\theta$ (From 7.(i))
6. $\neg\neg\theta$ (from (4) (5)).

(iii)

1. $\neg(\neg\neg\phi \to \phi)$ (hypothesis)
2. $\neg\neg\phi$ (from (1) and 7.(ii))
3. $\neg\phi$ (hypothesis)
4. $(\neg\neg\neg\phi \to \neg\phi) \to (\neg\neg\phi \to (\neg\phi \to \theta))$(from 6.(v))
5. $\neg\neg\neg\phi \to \neg\phi$ (Proposition 3)
6. $\neg\neg\phi \to (\neg\phi \to \theta)$ (from (4) above)
7. θ (from (2) and (3) above) and the result follows by the Deduction Theorem.

(iv)

1. $\neg\neg\theta \to \theta \vdash_{I1} \neg\phi \to (\phi \to \theta)$ (Proposition 6.(v))
2. $\neg(\neg\neg\theta \to \theta)$ (hypothesis)
3. $\neg\phi$(hypothesis)

4. ϕ (hy pothesis)
5. $\phi \to (\neg\neg\theta \to \phi)$ (Axiom 1)
6. $\neg\neg\theta \to \phi$ (from (4), (5), Modus Ponens)
7. $\neg\phi \to (\neg\neg\theta \to \neg\phi)$ (Axiom 1)
8. $\neg\neg\theta \to \neg\phi$ (from (3), (7), Modus Ponens)
9. $(\neg\neg\theta \to \neg\phi) \to ((\neg\neg\theta \to \phi) \to \neg\theta)$ (Axiom 3)
10. $(\neg\neg\theta \to \phi) \to \neg\theta$(from (8), (9), Modus Ponens)
11. $\neg\theta$ (from (6) and (10), Modus Ponens)
12. $\neg(\neg\neg\theta \to \theta) \vdash_{I^1} \neg\theta \to \theta$ (Proposition 7.(iii))
13. $\neg\theta \to \theta$ (from (2) and (12), Modus Ponens)
14. θ (from (11) and (13), Modus Ponens)
15. $\neg(\neg\neg\theta \to \theta) \vdash_{I^1} \neg\phi \to (\phi \to \theta)$ (from (2), (3), (4) and (14), Deduction Theorem)
16. $\neg\neg(\neg\phi \to (\phi \to \theta))$ (from (1) and (15), Proposition 6.(vi))
17. $\neg\phi \to (\phi \to \theta)$ (from (16) and Proposition 3 , Modus Ponens). ■

3. The Completeness Theorem for I^1

We take for granted all the usual definitions of logic matrices, evaluations, tautologies, etc. (as defined, for example, in [9]). Let

$$\mathcal{M} = \langle \{T, F^*, F\}, \{T\}, \neg, \to \rangle$$

be a logic matrix where T is the only distinguished logic value, and $\to, \neg$ are three-valued functions defined as follows:

$\to$	T	F^*	F
T	T	F	F
F^*	T	T	T
F	T	T	T

$\neg$	T	F^*	F
	F	F	T

We show, by a constructive method (which generalizes the well-known proof of Kálmar for the completeness of classical propositional logic) that I^1 is correct and complete with respect to the given logic matrix.

First we show a fundamental lemma:

PROPOSITION 8. *Let ϕ be a formula of I^1 and $q_1, q_2, \ldots, q_n$ its atomic components. Given an evaluation v, define for each q_i the following associate formulas*:

1. *if $v(q_i) = T$ then $q_i^+ = \neg\neg q_i \to q_i$ and $q_i^- = \neg\neg q_i$*
2. *if $v(q_i) = F$ then $q_i^+ = \neg\neg q_i \to q_i$ and $q_i^- = \neg q_i$*
3. *if $v(q_i) = F^*$ then $q_i^+ = \neg(\neg\neg q_i \to q_i)$ and $q_i^- = \neg\neg q_i$*

Let also $\phi' = \neg\neg\phi$ if $v(\phi) = T$,
$\phi' = \neg\phi$ if $v(\psi) = F$
$\phi' = \neg\neg\phi$ if $v(\phi) = F^*$
Then $q_1^+, q_1^-, q_2^+, q_2^- \ldots, q_n^+, q_n^- \vdash_{I^1} \phi'$

PROOF. By induction on the complexity of ϕ. In order to simplify notation, let the set $\{q_1^+, q_1^-, q_2^+, q_2^- \ldots, q_n^+, q_n^-\}$ be denoted by Δ.

(Case 1) If ϕ is atomic, then ϕ is q_i and the following derivations plus Proposition 2.(i) suffice for the proof:

subcase 1 For the case T, $\neg\neg q_i \to q_i, \neg\neg q_i \vdash_{I^1} \neg\neg q_i$

subcase 2 For the case F, $\neg\neg q_i \to q_i, \neg q_i \vdash_{I^1} \neg q_i$

subcase 3 For the case F*, $\neg(\neg\neg q_i \to q_i), \neg\neg q_i \vdash_{I^1} \neg\neg q_i$

(Case 2) If ϕ is $\neg\psi$ there are three subcases:

subcase 1 Suppose that $v(\psi) = T$; then ϕ' is $\neg\phi$,
and ψ' is $\neg\neg\psi$.
By the induction hypothesis,
$\Delta \vdash_{I^1} \neg\neg\psi$; but $\neg\neg\psi$ is ϕ', and we are done.

subcase 2 Suppose that $v(\psi) = F$; then ψ' is $\neg\psi$ and ϕ' is $\neg\neg\phi$. By the induction hypothesis
$\Delta \vdash_{I^1} \neg\psi$; but $\vdash_{I^1} \neg\psi \to \neg\neg\neg\psi$,
and thus
$\Delta \vdash_{I^1} \neg\neg\neg\psi$, i.e., $\Delta \vdash_{I^1} \neg\neg\phi$, but $\neg\neg\phi$ is ϕ'

subcase 3 Suppose that $v(\psi) = F^*$; then ψ must be an atomic formula hence ψ^- is $\neg\neg q_j$, for some j, and ϕ' is $\neg\neg q_j$. Thus $\neg\neg q_j \in \Delta$ and then $\Delta \vdash_{I^1} \neg\neg q_j$, but $\neg\neg q_j$ is ϕ'.

(Case 3) ϕ is $\psi \to \theta$. The argument now consists of four subcases:

subcase 1 Suppose that $v(\psi) = F$; then ψ' is $\neg\psi$ a nd ϕ' is $\neg\neg\phi$.From Proposition 7.(iv),$\vdash_{I1} \neg\psi \to (\psi \to \theta)$ By the induction hypothesis, $\Delta \vdash_{I1} \neg\psi$ and thus $\Delta \vdash_{I1} \psi \to \theta$. But $\vdash_{I1} (\psi \to \theta) \to \neg\neg(\psi \to \theta)$, thus $\Delta \vdash_{I1} \neg\neg(\psi \to \theta)$.

subcase 2 Suppose that $v(\theta) = T$; we consider first θ non-atomic. We have that θ' is $\neg\neg\theta$ and ϕ' is $\neg\neg\phi$, i.e., ϕ' is $\neg\neg(\psi \to \theta)$. But from Axiom 1, $\vdash_{I1} \neg\neg\theta \to (\psi \to \neg\neg\theta)$ and thus $\Delta \vdash_{I1} \psi \to \neg\neg\theta$, since $\Delta \vdash_{I1} \neg\neg\theta$,by hypothesis.

As θ is non-atomic, $\vdash_{I1} \neg\neg\theta \to \theta$. Hence , by transitivity, $\Delta \vdash_{I1} \psi \to \theta$ and by Proposition 5 $\Delta \vdash_{I1} \neg\neg(\psi \to \theta)$.

For θ atomic, say θ is q_j, then θ' is $\neg\neg q_j$, and ϕ' is $\neg\neg(\psi \to \theta)$. In this case, θ^+ is $\neg\neg q_j \to q_j$. By assumption, $\Delta \vdash_{I1} \neg\neg q_j$, hence $\Delta \vdash_{I1} q_j$ and consequently by Axiom 1 $\Delta \vdash_{I1} \psi \to q_j$, But by Proposition 5,$\Delta \vdash_{I1} \neg\neg(\psi \to q_j)$.

subcase 3 Suppose that $v(\psi) = T$ and $v(\theta) = F$; then ψ' is $\neg\neg\psi$, θ' is $\neg\theta$ and ϕ' is $\neg(\psi \to \theta)$.

Considering first ψ non-atomic, $\vdash_{I1} \neg\neg\psi \to \psi$ holds and from Proposition 6.(iv) we have:

$\neg\neg\psi \to \psi \vdash_{I1} (\psi \to (\neg\theta \to \neg(\psi \to \theta)))$

which allows us to deduce, using the assumptions $\Delta \vdash_{I1} \neg\neg\psi$ and

$\Delta \vdash_{I1} \neg\theta$,
$\Delta \vdash_{I1} \neg(\psi \to \theta)$i.e., $\Delta \vdash_{I1} \phi'$
If ψ is atomic, suppose q_j, then q_j^+ is $\neg\neg q_j \to q_j$ and ϕ' is $\neg(q_j \to \theta)$.
In this case, from 6.(iv) we have:
$\vdash_{I1} (\neg\neg q_j \to q_j) \to (q_j \to (\neg\theta \to \neg(q_j \to \theta)))$.
But by induction hypothesis $\Delta \vdash_{I1} q_j$and since $\Delta \vdash_{I1} \neg\theta$ one concludes $\Delta \vdash_{I1} \neg(q_j \to \theta)$, i.e., $\Delta \vdash_{I1} \phi'$.

subcase 4 Suppose that $v(\psi) = T$ and $v(\theta) = F^*$; In this case ϕ' is $\neg(\psi \to q_j)$.Also ψ' is $\neg\neg\psi$ and θ must be atomic, say q_j. Then θ^- is $\neg\neg q_j$ and θ^+ is $\neg(\neg\neg q_j \to q_j)$.
Let us consider first ψ atomic, say q_i; thus q_i^+ is $\neg\neg q_i \to q_i$ and q_i^- is $\neg\neg q_i$.
We have to show that $\Delta \vdash_{I1} \neg(q_i \to q_j)$ where $\Delta = \{\neg\neg q_i \to q_i, \neg\neg q_i, \neg(\neg\neg q_j \to q_j), \neg\neg q_j\}$. We have then:

(a) $\Delta, q_i \to q_j \vdash_{I1} q_j$(because $\Delta \vdash_{I1} q_i$)

(b) $\Delta, q_i \to q_j \vdash_{I^1} \neg\neg q_j \to q_j$ (from (a), Axiom 1)

(c) $\Delta, q_i \to q_j \vdash_{I^1} \neg(\neg\neg q_j \to q_j)$ (because $\neg(\neg\neg q_j \to q_j) \in \Delta$) hence

(d) $\Delta, q_i \to q_j \vdash_{I^1} \neg(q_i \to q_j)$ by Axiom 3 and Proposition 3, via the Deduction Theorem.

For ψ non-atomic $\vdash_{I^1} \neg\neg\psi \to \psi$ holds by Proposition 3; by hypothesis it also follows that
$\Delta \vdash_{I^1} \neg\neg\psi$
and the proof is similar to the one for ψ atomic. ■

PROPOSITION 9. (Completeness Lemma) *If an I^1-formula θ is a tautology then $\vdash_{I^1} \theta$*

PROOF. Denote by Δ_θ the set $q_1^+, q_1^-, q_2^+, q_2^- \ldots, q_n^+, q_n^-$ for all atomic components of θ. Let q be a particular atomic component of θ, and let $\Delta_0 = \Delta_\theta - \{q^+, q^-\}$. Consider three distinct evaluations v_1, v_2 and v_3 which differ in q, that is:

(a) $v_1(q) = T$; hence $q^+ = \neg\neg q \to q$, $q^- = \neg\neg q$.

(b) $v_2(q) = F^*$; hence $q^+ = \neg(\neg\neg q \to q)$,$q^- = \neg\neg q$.

(c) $v_3(q) = F$; hence $q^+ = \neg\neg q \to q$, $q^- = \neg q$.

Noting that θ is a tautology, from Proposition 8 we obtain:

1. from a, $\Delta_0, \neg\neg q \to q, \vdash_{I^1} \neg\neg q \to \theta$ and
2. from c, $\Delta_0, \neg\neg q \to q, \vdash_{I^1} \neg q \to \theta$
3. It follows from Proposition 6.(vi) that: $\neg\neg\neg q \to \neg q, \neg\neg q \to \theta, \neg q \to \theta \vdash_{I^1} \neg\neg\theta$
   But θ, being a tautology, cannot be atomic, and $\neg q$ is also non-atomic; thus by Proposition 3,
4. $\vdash_{I^1}, \neg\neg\theta \to \theta$ and
5. $\vdash_{I^1}, \neg\neg\neg q \to \neg q$ both hold. Hence we obtain:
6. $\Delta_0, \neg\neg q \to q \vdash_{I^1} \theta$
   On the other hand, from b
7. $\Delta_0, \neg(\neg\neg q \to q), \neg\neg q \vdash_{I^1} \theta$.
   Since by 7(ii)

8. $\neg(\neg\neg q \to q) \vdash_{I^1} \neg\neg q$, it follows that

9. $\Delta_0, \neg(\neg\neg q \to q) \vdash_{I^1} \theta$, which, together with (6) above, implies (using Theorem 6.(vi)):

10. $\Delta_0 \vdash_{I^1} \theta$

Repeating this procedure a finite number of times we obtain $\vdash_{I^1} \theta$. ■

PROPOSITION 10. (Completeness Theorem) *For any set of I^1-formulas $\Gamma, \{\theta\}$, $\Gamma \vdash_{I^1} \theta$ if and only if $\Gamma \models_{I^1} \theta$*

PROOF. It is sufficient (and easy) to check that the axiom schemes are I^1-tautologies and that the rule of Modus Ponens preserves tautologies. Thus, from Proposition 9, a formula is an I^1-tautology if and only if is derivable in I^1. The rest of the proof follows from the Deduction Theorem and the finiteness of the notion of proof. ■

4. Algebraization of I^1

In [2] Block and Pigozzi introduced a notion of algebraic semantics which directly generalizes the well-known definition of Lindenbaum-Tarski. Lewin, Mikemberg and Schwarze show in [8] that P^1 is algebraizable in the sense of Block and Pigozzi.

Following a similar approach, we give two distinct proofs (respectively, a syntactic and a semantic one) showing that I^1 is algebraizable in the above sense. For the sake of completeness, we recall the definitions involved in the proof.

Given a propositional language L, we denote by $\mathcal{F}(L)$ the set of well-formed formulas in L, defined in the usual way .

An *infere rule* (in L) is an ordered pair $\langle \Gamma, \phi \rangle$ where $\Gamma \subseteq \mathcal{F}(L)$ and $\phi \in \mathcal{F}(L)$.

A formula θ of L is said to be a *direct consequence* of $\Delta \subseteq \mathcal{F}(L)$ if there exists a substitution σ such that$\sigma(\phi) = \theta$ and $\sigma(\Delta) \subseteq \Delta$.

A *deductive system* Σ (over L), denoted by $\langle \mathcal{A}_\Sigma, \vdash_\Sigma \rangle$ consists of a set of axioms $\mathcal{A}_\Sigma \subseteq \mathcal{F}(L)$ and a collection of inference rules $\vdash_\Sigma \subseteq \mathcal{P}(\mathcal{F}(\mathcal{L})) \times \mathcal{F}(\mathcal{L})$ satisfying the following condition:

$\Lambda \vdash_\Sigma \psi$ iff ψ belongs to the smallest set of formulas which includes Λ and the instance of axioms of Σ, and is closed under the inference rules. The relations $\vdash_\Sigma$ are called the *inference rules of* Σ.An *L-algebra* is an algebra (in the sense, for example, of [9], that is, a set endowed with rations of the same similarity type as of L).

Let $\mathcal{K}$ a collection of L- algebras and let $\models_{\mathcal{K}}$ be a relation defined between a set of equations Γ and an equation $\phi \sim \psi$ in the following way:

$\Gamma \models_{\mathcal{K}} \phi \sim \psi$ iff for every algebra $\mathcal{A} \in \mathcal{K}$ and every interpretation s of the elements of the set $\Gamma \cup \{\phi \sim \psi\}$ as elements of $\mathcal{A}$ it holds that $\{s(\alpha) \sim s(\beta) : \alpha \sim \beta \in \Gamma\}$ implies $s(\phi) \sim s(\psi)$.

Let $\mathcal{I} = \langle \mathcal{A}, \vdash_I \rangle$ be a deductive system and $\mathcal{K}$ be a class of L-algebras; $\mathcal{K}$ is said to be an *alge braic semantics*for $\mathcal{I}$ if $\vdash_I$ can be interpreted by $\models_{\mathcal{K}}$ in the following sense: there exists a finite set of equations $\alpha_i(p) \sim \beta_i(p)$ $(1 \leq i \leq n)$ having p as the unique variable such that for every $\Gamma \cup \{\psi\}$ in $\mathcal{F}(L)$ the following condition holds:

$\Gamma \vdash_I \psi$ iff $\{\alpha_i(\theta) \sim \beta_i(\theta) : 1 \leq i \leq n, \theta \in \Gamma\} \models_{\mathcal{K}} \alpha_i(\psi) \sim \beta_i(\psi)$, for $1 \leq i \leq n$, where $\alpha_i(\mu)$ denotes the expression obtained by substitution of the variable p by μ in $\alpha_i(p)$.

The equations $\alpha_i(p) \sim \beta_i(p)$ are called *defining equations* of $\mathcal{I}$ and $\mathcal{K}$; we denote this system of equations by $\alpha(p) \sim \beta(p)$.

Given a deductive system $\mathcal{I}$ and $\mathcal{K}$ an algebraic semantics for $\mathcal{I}$ through defining equations $\alpha(p) \sim \beta(p)$, we say that $\mathcal{I}$ is *equivalent* to $\mathcal{K}$ if there exists a finite collection of formulas $\Delta_j(p,q)$ in two variables $(1 \leq i \leq m)$ such that for every equation $\phi \sim \psi$,

$$\phi \sim \psi = \models_{\mathcal{K}} \alpha(\phi \Delta \psi) \sim \beta(\phi \Delta \psi)$$

where $\phi \Delta \psi$ abbreviates $\phi \Delta_i \psi$ for $1 \leq i \leq m$.

The system of formulas Δ satisfying the condition above is called a *system of equivalence formulas* for $\mathcal{I}$ and $\mathcal{K}$.

The proof of the following results (Propositions 11 to 14) can be found in [?].

PROPOSITION 11. *Let $\mathcal{K}$ be an algebraic semantics for $\mathcal{K}$; if Δ is a system of equivalent formulas then for every set of equations in Γ and every equation $\phi \sim \psi$, we have:*

(i) $\Gamma \models_{\mathcal{K}} \phi \sim \psi$ iff $\{\rho \Delta \sigma : \rho \sim \sigma \in \Gamma\} \vdash_I \phi \Delta \psi$

((ii) $\mu \dashv\vdash \delta(\mu) \Delta \epsilon(\mu)$ for every $\mu \in \mathcal{F}(L)$ and $(\delta \sim \epsilon) \in \Delta$.

Conversely, if there exists a system Δ of formulas satisfying the above conditions then $\mathcal{K}$ is an algebraic semantics for $\mathcal{I}$ having Δ as a system of equivalence formulas.

A deductive system $\mathcal{I}$ is *algebrable* if $\mathcal{I}$ admits an equivalent algebraic semantics.

It is interesting to remark that if $\mathcal{I}$ admits an equivalent algebraic semantics $\mathcal{K}$ then it follows from the definition that $\vdash_I$ is interpretable in $\models_{\mathcal{K}}$, Proposition 11.(i) means that $\models_{\mathcal{K}}$ is interpretable in $\vdash_I$, and finally Proposition 11.(ii) means that, in a certain sense, such interpretations are mutually inverse.

PROPOSITION 12. *Let $\mathcal{I}$ be a deductive system given by a set of axioms $\mathcal{A}_\Sigma$ and by a collection of inference rules $\vdash_I$. Suppose that $\mathcal{I}$ is algebraizable having Δ as a set of equivalence formulas and defining equations $\delta \sim \epsilon$. Under these conditions the only quasi-variety which is an algebraic semantics equivalent for $\mathcal{I}$ is axiomatized by the identities:*

1. $\delta(\phi) \sim \epsilon(\phi)$ *for every* $\phi \in \mathcal{A}_\Sigma$
2. $\delta(p\Delta p) \sim \epsilon(p\Delta p)$ *and by the quasi-identities*
3. $((\delta(\psi_1) \sim \epsilon(\psi_1) \wedge \ldots \wedge \delta(\psi_n) \sim \epsilon(\psi_n)) \rightarrow \delta(\phi) \sim \epsilon(\phi)$ *for every* $\langle\{\psi_1, \ldots, \psi_n\}, \phi\rangle \in \vdash_I$
4. $\delta(p\Delta q) \sim \epsilon(p\Delta q) \rightarrow p \sim q$.

The following propositions give, respectively, the syntactic and the semantic characterization of an algebraizable deductive system:

PROPOSITION 13. *A deductive system $\mathcal{I}$ is algebraizable if and only if there exists a set Δ of formulas in two variables and a system of equations $\delta \sim \epsilon$ in one variable such that the following conditions are verified for all ϕ, ψ and θ in $\mathcal{F}(L)$:*

1. $\phi\Delta\phi$
2. $\phi\Delta\psi \vdash_I \psi\Delta\phi$
3. $\phi\Delta\psi, \psi\Delta\theta \vdash_I \phi\Delta\theta$
   (*that is, Δ is an equivalence relation on $\mathcal{F}(L)$*)
4. *for every n-ary connective ω and formulas $\phi_1, \ldots, \phi_n, \psi_1, \ldots, \psi_n$ in $\mathcal{F}(L)$):* $\phi_1\Delta\psi_1, \ldots, \phi_n\Delta\psi_n \vdash_I \omega(\phi_1, \ldots, \phi_n)\Delta\omega(\psi_1, \ldots, \psi_n)$ (*that is, Δ is a congruence relation*)
5. $\theta \dashv\vdash_I \delta(\theta)\Delta\epsilon(\theta)$

In this case, Δ is a set of equivalence formulas and $\delta \sim \epsilon$ are defining equations.

An *L-matrix* is a pair $\mathcal{M} = \langle \mathcal{A}, D \rangle$ where $\mathcal{A}$ is an algebra with universe A and $D \subseteq A$ is set of *distinguished elements.*

A class M of L-matrices *determines* a deductive system $\mathcal{I}_{\mathrm{M}} = \langle L, \vdash_{\mathrm{M}} \rangle$ in the following way:

$\Gamma \vdash_{\mathrm{M}} \phi$ if and only if for every $\mathcal{M}$ in M and for every interpretation s of the variables in $\Gamma \cup \{\phi\}$ on $\mathcal{A}$, if $s(\rho) \in D$ for all $\rho \in \Gamma$ then $s(\phi) \in D$.

PROPOSITION 14. *Let* M *be a class of matrices and* $\mathcal{I}_{\mathrm{M}}$ *a deductive system defined by* M. *Then* $\mathcal{I}_{\mathrm{M}}$ *is algebraizable iff there exist unary formulas* δ_i, ϵ_i *and binary formulas* Δ_i $(1 \leq i \leq m)$ *such that for every matrix* $\mathcal{M} = \langle \mathcal{A}, D \rangle$ *in* M *the relation*

$$R(\mathcal{M}, \Delta) = \{(a, b) \in \mathcal{A} \times \mathcal{A} : \Delta_j^{\mathcal{A}}(a, b) \in D \text{ for } 1 \leq j \leq m\}$$

verifies the conditions:

1. $R(\mathcal{M}, \Delta)$ *is a congruence relation on* $\mathcal{A}$
2. *for every* $a \in A$, $a \in D$ *iff* $(\delta_i(a), \epsilon_i(a)) \in R(\mathcal{M}, \Delta)$ *for* $1 \leq i \leq m$

We show, in the sequel, the algebraization of I^1 in the sense discussed here:

PROPOSITION 15. (Syntactic form) I^1 *is algebraizable.*

PROOF. We give first a synctactic proof. Denote, for $\phi, \psi \in \mathcal{F}(L)$: $\delta(\phi) = (\phi \to \phi) \to \phi$, $\epsilon(\phi) = \phi \to \phi$,$\phi\Delta_1\psi = \phi \to \psi$, $\phi\Delta_2\psi = \psi \to \phi$,$\phi\Delta_3\psi = \neg\phi \to \neg\psi$, $\phi\Delta_4\psi = \neg\psi \to \neg\phi$.

It is sufficient to show that Δ, ϕ, ψ satisfy conditions(1) - (5) of Proposition 13:

1. $\phi\Delta\phi$ follows from Proposition 2;
2. $\phi\Delta\psi \vdash_I \psi\Delta\phi$ follows from the definition of Δ;
3. $\phi\Delta\psi, \psi\Delta\theta \vdash_I \phi\Delta\theta$ follows from Proposition 2 (i.e., Δ is an equivalence relation on $\mathcal{F}(L)$)
   (a) $\phi\Delta\psi \vdash_I \neg\phi\Delta\neg\psi$, follows from the axioms of I^1 and the definition of Δ;
   (b) $\phi_1\Delta\psi_1, \phi_2\Delta\psi_2 \vdash_I (\phi_1 \to \phi_2)\Delta(\psi_1 \to \psi_2)$, that is, it has to be proved that:
      (i) $\phi_1\Delta\psi_1, \phi_2\Delta\psi_2 \vdash_I (\phi_1 \to \phi_2) \to (\psi_1 \to \psi_2)$

(ii) $\phi_1 \Delta \psi_1, \phi_2 \Delta \psi_2 \vdash_I (\psi_1 \to \psi_2) \to (\phi_1 \to \phi_2)$
(iii) $\phi_1 \Delta \psi_1, \phi_2 \Delta \psi_2 \vdash_I \neg(\phi_1 \to \phi_2) \to \neg(\psi_1 \to \psi_2)$
(iv) $\phi_1 \Delta \psi_1, \phi_2 \Delta \psi_2 \vdash_I \neg(\psi_1 \to \psi_2) \to \neg(\phi_1 \to \phi_2)$

but this follows easily from elementary syntactic properties of I^1;hence Δ is a congruence relation.

4. $\theta \dashv\vdash \delta(\theta)\Delta\epsilon(\theta)$,
that is, $\theta \dashv\vdash ((\theta \to \theta) \to \theta)\Delta(\theta \to \theta)$, or, in more details,
(a) $((\theta \to \theta) \to \theta)\Delta(\theta \to \theta) \vdash_I \theta$
and
(b) $\theta \vdash_I ((\theta \to \theta) \to \theta)\Delta(\theta \to \theta)$
Part (a) is obtained considering that $(\theta \to \theta) \to ((\theta \to \theta) \to \theta) \in \Delta$ and $\vdash_I (\theta \to \theta)$
For part (b), is is sufficient to prove:

(a) $\theta \vdash_I ((\theta \to \theta) \to \theta) \to (\theta \to \theta)$
(b) $\theta \vdash_I (\theta \to \theta) \to ((\theta \to \theta) \to \theta)$
(c) $\theta \vdash_I \neg((\theta \to \theta) \to \theta) \to \neg(\theta \to \theta)$
(d) $\theta \vdash_I \neg(\theta \to \theta) \to \neg((\theta \to \theta) \to \theta)$

These easy derivations are left to the reader. ■

According to Proposition 14, we also give a semantic proof for the aforementioned algebraization:

PROPOSITION 16. *The algebraic semantics associated with I^1 is thequasi-variety generated by the three-element algebra M which is the characteristic matrix of I^1.*

PROOF. Let Δ, δ, ϵ be as in the previous proof; we show that these formulas satisfy conditions (1) and (2) of Proposition 14:

Due to the Completeness Theorem, I^1 is semantically defined by M = $\{\mathcal{M}\}$. We have to prove that:

1. $R(\mathcal{M}, \Delta)$ is a congruence on $\{T, F^*, F\}$
2. for $x \in \{T, F^*, F\}$, $x \in \{T\}$ iff $(\delta(x), \epsilon(x)) \in R(\mathcal{M}, \Delta)$

For the first part, observe that:
$(T, F) \notin R(\mathcal{M}, \Delta)$ since $T\Delta_1 F = F$
$(T, F^*) \notin R(\mathcal{M}, \Delta)$ since $T\Delta_1 F^* = F$

$(F^*, T) \notin R(\mathcal{M}, \Delta)$ since $F^* \Delta_2 T = F$
$(F^*, F) \notin R(\mathcal{M}, \Delta)$ since $F^* \Delta_4 F = F$
$(F, T) \notin R(\mathcal{M}, \Delta)$ since $F \Delta_2 T = F$
$(F, F^*) \notin R(\mathcal{M}, \Delta)$ since $F \Delta_3 F^* = F$

Therefore $R(\mathcal{M}, \Delta)$ must be the identity, hence a congruence relation.

For the second part, we have:

$\delta(T) = (T \to T) \to T = T \to T = T = \epsilon(T)$
$\delta(F) = (F \to F) \to F = T \to F = F \neq \epsilon(F)$
$\delta(F^*) = (F^* \to F^*) \to F^* = T \to F^* = F \neq \epsilon(F^*)$
but
$(\delta(F), \epsilon(F)) \notin R(\mathcal{M}, \Delta)$ and
$(\delta(F^*), \epsilon(F^*)) \notin R(\mathcal{M}, \Delta)$, thus
$x \in \{T\}$ iff $(\delta(x), \epsilon(x)) \in R(\mathcal{M}, \Delta)$. ■

5. The maximality of I^1

In this section we show the intuitionistic character of the calculus I^1 and prove that it is maximal in a sense to be made precise. Firstly, conjunction $\wedge$ and disjunction $\vee$ can be defined on I^1 as:

$\alpha \wedge \beta = \neg(((\alpha \to \alpha) \to \alpha) \to \neg((\beta \to \beta) \to \beta))$
and
$\alpha \vee \beta = (\neg(\beta \to \beta) \to \beta) \to ((\alpha \to \alpha) \to \alpha)$
giving, respectively, the tables:

$\wedge$	T	F^*	F
T	T	F	F
F^*	F	F	F
F	F	F	F

$\vee$	T	F^*	F
T	T	T	T
F^*	T	F	F
F	T	F	F

With such definitions, with the help of the Completeness Theorem it can easily be checked that I^1 is weakly-intuitionistic (and, moreover, that the law of exluded middle is false for non-atomic formulas only).

For the maximality of I^1, let us call any wff schema (in the language of I^1) which is a classical tautology but not an I^1-tautology a *paralogism*. We thus have:

PROPOSITION 17 *Let α be a paralogism a nd I^{1+} be obtained by adding α to I^1; then I^{1+} collapses to PC.*

PROOF. We need to show that, under the given conditions, I^{1+} and PC prove the same theorems. Let α be as above and $\{P_1, \ldots, P_m\}$ be metavariables for atomic components.

Clearly, there exists an I^1- valuation such that $v(\alpha) \neq T$ and $v(P_{i_k}) = F^*$ for some $1 \leq i_k \leq m$ (such v must exist, for restricting v to $\{T, F\}$ produces a PC-tautology).

Let α' be the wff obtained from α substituting P_i by $\neg\neg P_i \rightarrow P_i$ if $v(P_i) \neq F^*$, and by $\neg(\neg\neg P_i \rightarrow P_i)$ if $v(P_i) = F^*$.

Thus, evey valuation which assigns F^* to P_{i_k} also assigns F to α', and hence, noting that the defined connective $\vee$ has the appropriate properties, we see that the following is an I^1-tautology:

$$\models_{I^1} \alpha' \rightarrow (\neg\neg P_{i_1} \rightarrow P_{i_1} \vee \ldots \vee \neg\neg P_{i_m} \rightarrow P_{i_m})$$

In virtue of the Completeness Theorem, this formula schema can be deduced in I^1, and consequently $\vdash_{I^{1+}} (\neg\neg P_{i_1} \rightarrow P_{i_1} \vee \ldots \vee \neg\neg P_{i_m} \rightarrow P_{i_m})$.

In particular, $\vdash_{I^{1+}} (\neg\neg P_{i_1} \rightarrow P_{i_1} \vee \ldots \vee \neg\neg P_{i_1} \rightarrow P_{i_1})$; considering that $\vdash_{I^{1+}} A \vee A \rightarrow A$, one deduces:

$\vdash_{I^{1+}} \neg\neg P_{i_1} \rightarrow P_{i_1}$, having as instances all propositional variables. By Proposition 6, all PC tautologies can bededuced in I^{1+}. ■

6. The Hierarchies P^n and I^n

It is a fact that most of the interesting multivalued logics are indeed three-valued logics; though apparently very simple, three-valued systems like P^1 and I^1 can reveal surprisingly interesting properties when investigated from the appropriate viewpoint.

Three-valued logics occur as the first natural candidates for genuine *mathematical experimentation* with truth-values, examples of which are the three-valued algebras for I^1 presented in this paper and for P^1 given in [8].

From a more philosophical viewpoint, even accepting, the view that has been put forward by some authors (see for example [10]) that it is not easy to assign philosophically sensible meanings to n-valued logics in general, this difficulty does not seem to be insurmountable in the particular case of three-valued logics. As argued by Haack in [6], many-valued logics do not *necessarily* call for the rejection of bivalence. Additional truth-values, instead of representing elements extaneous to the discussion, can be seen as refinements of the same concept of bivalence.

Guided by this idea, we propose multivalued generalizations of the systems P^1 and I^1 , which we call P^n and I^n. These calculi are subsystems of classical logic and belong to more complex hierarchies of finite-valued paraconsistent and weakly-intuitionistic logics (also called *paracomplete* logics, cf. [1]), a not much studied class of calculi of which few examples are known. Though every multivalued logic can be fully characterized through the method of finitely generated trees (cf.for example [4]) this is not the case for paraconsistent logics: the hierarchy of calculi C_n, even if modified so as to allow reduction of negations *cannot* be characterised by finite tables (see [5]). As is well known, a similar phenomenon occurs with intuitionistic logic. Paraconsistent and intuitionistic deductions, indeed, behave in paradigmatic opposite but dual ways, perhaps the more intuitively appealing being the fact that paraconsistent deductionis extremely liberal and intuitionistic deduction is extremely conservative.

Several different hierarchies can be defined; the one we propose is situated between such extremes. Studying these hierarchies, we believe, can shed some light on the phenomenon of paraconsistency and its relation to classical and intuitionistic logics.

The calculi P^n (respectively, I^n) are designed to form a hierarchy of logics such that :

1. P^n and I^n, for $n > 1$, are $(n+2)$-valued logics; in particular, P^0 and I^0 coincide with classical propositional logic.

2. P^{n+1} is a maximal subsystem of P^n, and I^{n+1} is a maximal subsystem of I^n (where the notion of maximality is the one introduced in the previous section).

The branch P^n of this hierarchy will be considered first. The language, common to all systems, is defined as usual, starting from implication $\rightarrow$ and negation $\neg$. The truth-values of P^n are the members of the set

$$L_n = \{T, T^*, T^{**}, \ldots, F\}$$

of which F is the only non-distinguished value.

For example, the matrices for P^2 are the following:

$\rightarrow$	T	T^*	T^{**}	F
T	T	T	T	F
T^*	T	T	T^*	F
T^{**}	T	T^*	T	F
F	T	T	T	T

$\neg$	T	T^*	T^{**}	F
	F	T	T^*	T

For I^2, the truth-values are the membersof the set $L_n = \{F, F^*, F^{**}, \ldots, T\}$ of which T is the only distinguished value. The matrices for I^2 are:

$\rightarrow$	T	F^*	F^{**}	F
T	T	F	F	F
F^*	T	T	F^*	T
F^{**}	T	F^*	T	T
F	T	T	T	T

$\neg$	T	F^*	F^{**}	F
	F	T	F^*	T

Adapting our detailled proofs for I^1 , it can be proved that P^2 and I^2 are correct and complete with respect to appopriate sets of axioms (the only rule of inference being *Modus Ponens* in all cases). However, no uniform proof of completeness for the whole hierarchy has yet been found.

7. Concluding Remarks

From the purely logical viewpoint, an interesting question is to study the inductive limits of the classes P^n and I^n : which system does one obtain and what are their algebraic properties?

From the algebraic side, it seems surprising that such distinct (indeed, dual) systems as P^1 and I^1 can be proved to be algebraizable by the same sets Δ and equations $\delta \sim \epsilon$.

Although in [2] an example is given of two logics be algebraizable using the same set Δ, distinct defining equations are necessary. But the case of P^1 and I^1 naturally poses the question: what are the relations between two distinct systems so that their corresponding algebraizations admit the same Δ and the same equations $\delta \sim \epsilon$?

Duality (in this case between the intuitively opposite concepts of paraconsistent deduction and of intuitionistic deduction) seems to play a strong role; it seems promising to define precisely such a concept in generality and explore its consequences. In particular, we conjecture that, as a consequence of duality, I^1 could be shown not to be algebraizable in the strict sense of Rasiowa [9].

References

[1] A. I. Arruda, A survey of paraconsistent logic, in: A.I. Arruda, N. C. A. da Costa, and R Chuaqui, editors, *Mathematical logic in Latin America*, pages 1–41. North-Holland, 1980.

[2] W. J. Block and D. Pigozzi, Algebraizable logics, *Memoirs of the American Mathematical Society*, 77(396), 1989.

[3] Suzana Laino Cândido, Sobre a lógica não-alética, Master's thesis, Pontifícia Universidade de São Paulo, São Paulo, Brazil, 1992.

[4] Walter A. Carnielli, Systematization of the finite many-valued logics through the method of tableaux, *The Journal of Symbolic Logic*, 52:473–493, 1987.

[5] Walter A. Carnielli, Many-valued logics and plausible reasoning, in: *Proceedings of Intl. Symp. on Multiple-valued Logic*, Charlotte, 1990. IEEE.

[6] S. Haack, *Philosophy of logics*, Cambridge University Press, 1978.

[7] Andrea Loparič and Newton C. A. da Costa, Paraconsistency, paracompleteness and induction, *Logique et Analyse*, 113:73–80, 1986.

[8] R.A. Lewin, I. F. Mikenberg, and M. G. Schwarze, Algebraization of paraconsistent logic p^1, *The Journal of Non-Classical Logic*, 7(2):1 45–154, 1990.

[9] H. Rasiowa, *An Algebraic Approach to Non-Classical Logics*, North-Holland Publ. Co., Amsterdam, 1974.

[10] Dana Scott, Does many-valued logic have any use?in: S.Körner, editor, *Philosophy of Logic*. Blackwell, Oxford, 1976.

[11] A. M. Sette, On the propositional calculus p^1, *Mathematic Japonicæ*, 18(13):173–180, 1973.

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